

Project Initialization and Planning Phase

Date	05 July 2026
Team ID	
Project Title	Smart Lender
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To automate the bank loan eligibility evaluation process using machine learning to make instant, data-driven decisions..
Scope	Includes data cleaning, SMOTE balancing, training 5 ML algorithms, deploying a tuned XGBoost model, database logging, and building a Flask dashboard UI.
Problem Statement	
Description	Manual loan underwriting is slow, paper-heavy, highly subjective, and prone to human error and personal decision-making bias.
Impact	Causes long processing backlogs, customer drop-offs to faster competitors, and increased default risks due to poor risk evaluation.
Proposed Solution	
Approach	Wrap a tuned XGBoost model inside a Flask web application to process and predict loan eligibility instantly when parameters are submitted.
Key Features	Interactive 3-step form wizard, automatic input scaling, real-time approval probability scores, and local SQLite log tracking with delete options.

--	--

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv